ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

TECHNICAL PROGRAMMING

PAPER 1 · PRACTICE C

MARKS: 120 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Laid out like the DHET first paper (120 marks, Python). Section A (40): multiple choice with a pseudocode trace, true or false, choosing the correct term and matching · Section B (80): computer systems, software types and the program development life cycle, a vowel-counting program, Git, a robot-parts list program to study, dictionary coding, and classes and objects.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. All program code is written in Python 3. Code is not marked strictly on syntax.
6. Write neatly and legibly.

SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 Which type of memory loses its contents when the power is switched off?

A ROM B Flash memory C RAM D SSD

1.2 What is the main job of an operating system?

A To manage the hardware and resources of the computer
B To edit photographs
C To browse the internet
D To write program code

1.3 Which ONE of the following is free, open-source software?

A Microsoft Office
B Adobe Photoshop
C Windows 11
D LibreOffice

1.4 Which Python data type stores key–value pairs?

A list B dictionary C tuple D string

1.5 A short description of a feature, written from the user's point of view, is called a …

A user story.
B test case.
C flowchart.
D bug report.

1.6 Which control structure repeats a group of statements?

A Sequence B Selection C Iteration D Declaration

1.7 Which protocol is used to open a secure command-line session on a Raspberry Pi from another computer?

A VNC B SSH C FTP D HTTP

1.8 What does len([3, 7, 9]) return?

A 7 B 9 C 19 D 3

1.9 Which processor architecture is used in most smartphones and in the Raspberry Pi?

A ARM B x86 C PowerPC D SPARC

1.10 Study the pseudocode below. What is displayed?

```
Total ← 0
FOR k ← 1 TO 3
    Total ← Total + k * 2
NEXT k
DISPLAY Total
```

A 6 B 12 C 9 D 18

(10)

[10]

QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Write only 'True' or 'False' next to the question number (2.1–2.10) in the ANSWER BOOK.

2.1 A Python tuple can be changed after it has been created.

2.2 VNC lets you see and control the desktop of another computer.

2.3 A function can return a value to the code that called it.

2.4 The binary number 1001 is equal to the decimal number 10.

2.5 Git keeps a history of the changes made to files.

2.6 The first item of a Python list has index 1.

2.7 An interpreter runs a program one statement at a time.

2.8 A while loop always runs at least once.

2.9 Comments are ignored by the Python interpreter.

2.10 Pseudocode helps a programmer to plan the logic before writing code.

(10)

[10]

QUESTION 3

Choose the correct term from those in brackets and write it next to the question number (3.1–3.10) in the ANSWER BOOK.

3.1 A solid-state drive stores data in (volatile/non-volatile) memory.

3.2 Running several programs at the same time is called (multitasking/multiplexing).

3.3 The PDLC phase in which the program is checked for errors is (analysis/design/testing).

3.4 A diagram that shows the steps of an algorithm with symbols is a (flowchart/storyboard/user story).

3.5 Choosing between two paths with if … else is an example of (sequence/selection/iteration).

3.6 The items of a Python list are written inside (round brackets/square brackets/curly braces).

3.7 Data sent over an SSH connection is (encrypted/unencrypted).

3.8 The official operating system of the Raspberry Pi is (Raspberry Pi OS/Windows 11/macOS).

3.9 The Git command that sends local commits to GitHub is git (push/pull/clone).

3.10 An app that works without an internet connection is a (standalone/web/cloud) app.

(10)

[10]

QUESTION 4

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–O) next to the question number (4.1–4.10) in the ANSWER BOOK.

	COLUMN A		**COLUMN B**
4.1	Removes and returns the last item of a list	A	attribute
4.2	Translates the whole program into machine code before it runs	B	IPO chart
4.3	A blueprint from which objects are created	C	compiler
4.4	A variable that belongs to an object	D	nano
4.5	A series of sketches showing screens and how the user moves between them	E	pop()
4.6	A text editor that runs in the Raspberry Pi terminal	F	storyboard
4.7	An error that occurs while a program is running	G	class
4.8	A data structure in which each value is found by its key	H	dictionary
4.9	A table that shows the inputs, the processing and the outputs	I	web application
4.10	Software that is used through a web browser	J	exception
		K	tuple
		L	method
		M	SSH
		N	interpreter
		O	append()

(10)

[10]

TOTAL SECTION A: [40]

SECTION B

QUESTION 5

5.1 Explain why a computer with more RAM can run several programs at the same time more smoothly. (2)

5.2 Compare standalone (desktop) software and web applications. Copy the table below into your answer book and complete it. (6)

Aspect	Standalone software	Web application
Definition		
Platform dependency		
Internet requirement		

5.3 Answer the following questions about the program development life cycle (PDLC):

5.3.1 Name, in order, SIX phases of the PDLC. (3)

5.3.2 Differentiate between the analysis phase and the design phase. (2)

5.3.3 Define the term debugging. (2)

5.4 Study the Python program below and answer the questions that follow.

```
1  word = input("Enter a word: ")
2  count = 0
3  for letter in word:
4      if letter in "aeiou":
5          count += 1
6  print(word.upper(), "has", count, "vowels")
```

5.4.1 What does line 4 check? (1)

5.4.2 What is displayed if the user enters robot? (2)

5.5 Give TWO reasons why ARM processors are popular in robots and other battery-powered devices. (2)

5.6 Write the Git command that copies a repository from GitHub onto your computer. (1)

5.7 Explain the purpose of the git log command. (2)

[23]

QUESTION 6

Study the following program code and answer the questions.

```
 1  parts = ["motor", "sensor", "battery", "wheel"]
 2
 3  def show_parts():
 4      print("\nParts in stock:")
 5      for number, part in enumerate(parts, start=1):
 6          print(number, part)
 7
 8  def add_part():
 9      part = input("Part to add: ").strip().lower()
10      if part == "":
```

```
11          print("Nothing added.")
12      elif part in parts:
13          print(part, "is already in the list.")
14      else:
15          parts.append(part)
16
17  def use_part():
18      part = input("Part used: ").strip().lower()
19      if part in parts:
20          parts.remove(part)
21          print(part, "removed.")
22      else:
23          print(part, "is not in stock.")
24
25  while True:
26      choice = input("\n1 Show  2 Add  3 Use  4 Quit: ")
27      if choice == "1":
28          show_parts()
29      elif choice == "2":
30          add_part()
31      elif choice == "3":
32          use_part()
33      elif choice == "4":
34          break
35      else:
36          print("Choose 1 to 4.")
```

6.1 What does enumerate(parts, start=1) do in line 5? (2)

6.2 Explain what .strip().lower() does in line 9 and why it is useful. (2)

6.3 What happens if the user chooses option 2 and types Motor? (2)

6.4 What does the keyword elif do? (1)

6.5 The user adds "gripper" (option 2), then uses "battery" (option 3), then chooses option 1. Write down the exact output of show_parts(). (4)

6.6 Write a function named count_parts() that displays how many parts are in stock, using len(). (4)

6.7 How does the program stop running? (2)

[17]

QUESTION 7

7.1 Write Python code to create a dictionary named battery_levels with the following four key–value pairs, where the keys are robot names and the values are battery percentages: (4)

```
'R1': 85,
'R2': 40,
'R3': 15,
'R4': 60
```

7.2 Write a single line of Python code that displays the battery level of R3 with an informative message. (2)

7.3 Write Python code that asks the user for a new robot's name and battery level, and adds this key–value pair to battery_levels. (4)

7.4 Write Python code that asks the user for the name of a robot that has been taken out of service. If the robot is in the dictionary, remove it and display a message; if not, display "Robot not found". (8)

7.5 Write Python code that uses a for loop to display each robot's name and the word "CHARGE" if its level is below 20, or "OK" otherwise. (4)

[22]

QUESTION 8

8.1 Define the term encapsulation. (2)

8.2 Differentiate between a class and an object. Copy the table below into your answer book and complete it. (6)

Aspect	Class	Object
Definition		
Purpose		
Example		

8.3 Write the code for a Python class named Sensor with the attributes name, pin and value. Include a constructor that sets name and pin from its parameters and sets value to 0. (7)

8.4 Referring to QUESTION 8.3, write the code to create a Sensor object named temp1 with the name "Temperature" and pin 4, and then display its name. (3)

[18]

TOTAL SECTION B: [80]

TOTAL: 120

END OF QUESTION PAPER